

Application Architecture -- Guided Notes ANSWER KEY

Unit tp-4.2 | CompTIA Network+ N10-009 Obj. FC0-U71 4.2 | N10-008 FC0-U71 4.2

For instructor use / distribute after students complete the notes.

Question 1

In a _____ -tier architecture, the user interface, business logic, and data are all in one application layer. In a _____ -tier architecture, the presentation and data tiers are separated. A _____ -tier (or n-tier) architecture separates presentation, application logic, and data into distinct layers on different servers.

Answer: Single (1); two (2); three (3).

Question 2

The _____ tier handles what the user sees -- HTML, CSS, JavaScript in a browser. The _____ (logic) tier processes business rules and application logic (e.g., a web server running Python or Java). The _____ tier stores persistent data in a database. Each tier communicates with the tier _____ it.

Answer: Presentation; application (middle/business logic); data; adjacent to (above or below).

Question 3

In a _____ architecture, all functionality is packaged in one deployable unit. It is simpler to _____ initially but becomes harder to _____ as the application grows. In a _____ architecture, the application is split into small, independently deployable services that communicate over APIs.

Answer: Monolithic; develop; scale/maintain; microservices.

Question 4

Serverless computing means the developer writes code (called _____) without managing servers at all. The cloud provider runs the function on demand and charges per _____. It scales _____ and costs nothing when not running. Examples include AWS _____ and Azure _____.

Answer: Functions; invocation (execution); automatically; Lambda; Functions.

Question 5

A _____ load balancer distributes incoming network requests across multiple backend servers to prevent any one server from being _____ and to improve availability. If one server goes down, the load balancer _____ traffic to the remaining healthy servers.

Answer: Load balancer; overloaded (overwhelmed); redirects.

Question 6

A cache is a temporary storage location that holds frequently accessed data to reduce _____ to the primary data source. A CDN (Content Delivery Network) caches _____ at edge servers geographically close to users. Redis and Memcached are examples of _____ caching software used in web backends.

Answer: Requests (load); static content (images, files); in-memory.

Question 7

High availability (HA) means a system is designed to minimize _____ time. It is measured as a percentage of uptime (e.g., _____ -- 'five nines' -- means 99.999% uptime, or less than _____ minutes of downtime per year). HA is achieved through _____ (no single point of failure) and failover systems.

Answer: Downtime; 99.999%; 5.26 minutes; redundancy.

Question 8 -- Real World Application

REAL WORLD: A retail website runs on a single web server. On Black Friday their traffic spikes 20x and the server crashes, causing 3 hours of downtime and \$200,000 in lost sales. Describe two architectural changes that would prevent this outcome and explain how each helps.

Answer: Change 1 -- Add a load balancer with multiple web server instances: distribute traffic across 3-5 web servers behind a load balancer. When traffic spikes 20x, the load is spread across all servers. If one fails, the others continue serving traffic. Auto-scaling groups (available in cloud providers) can automatically add new server instances when CPU/traffic thresholds are exceeded, handling unpredictable demand spikes. Change 2 -- Use a CDN (Content Delivery Network): static assets (product images, CSS, JavaScript) represent a large portion of web traffic. Moving these to a CDN means millions of requests for static files never hit the origin web server at all -- they are served from edge nodes. This dramatically reduces load on the backend during traffic spikes.